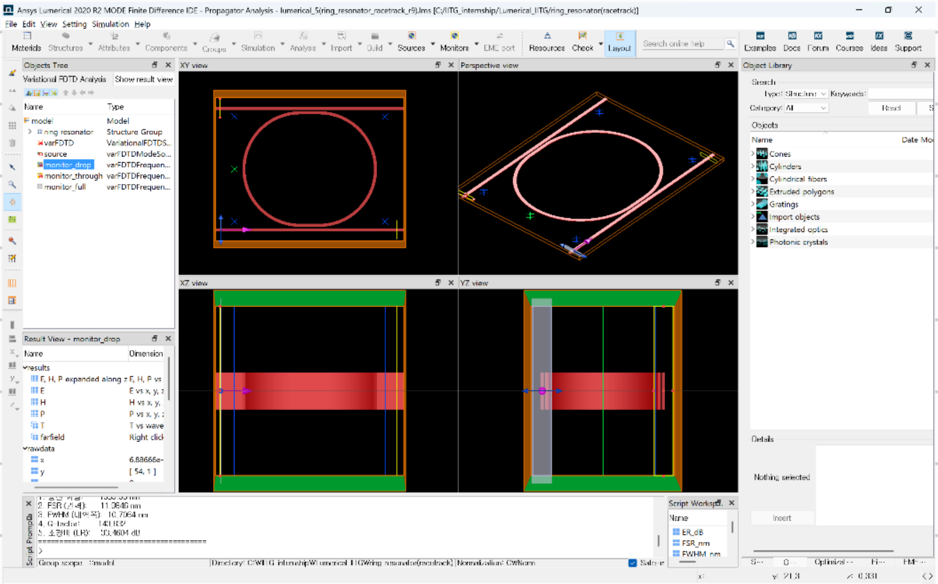
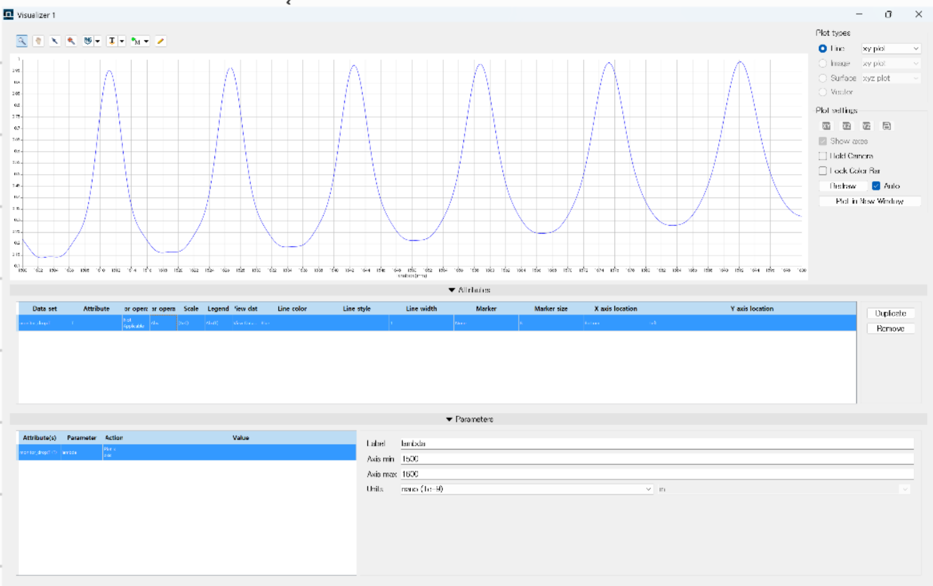


radius 값을 8~12μm로 변화하며 FSR, FWHM 값 확인하기

1) radius 8μm

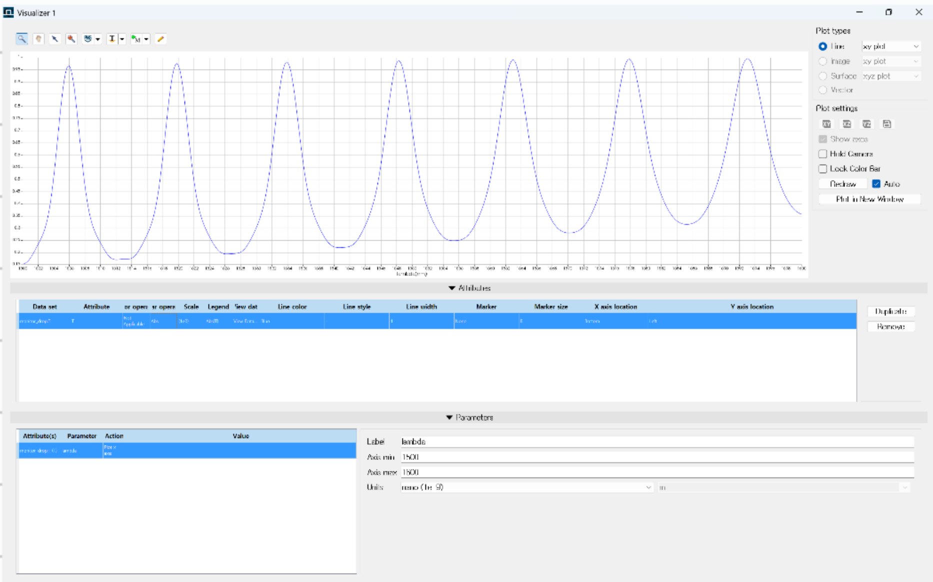


----

1. Resonant Wavelength (nm): 1592.13
2. FSR (nm): 16.9621
3. FWHM (nm): 12.2913
4. Q-factor: 129.533
5. Extinction Ratio (dB): 8.53955

===================================

2) radius 9μm

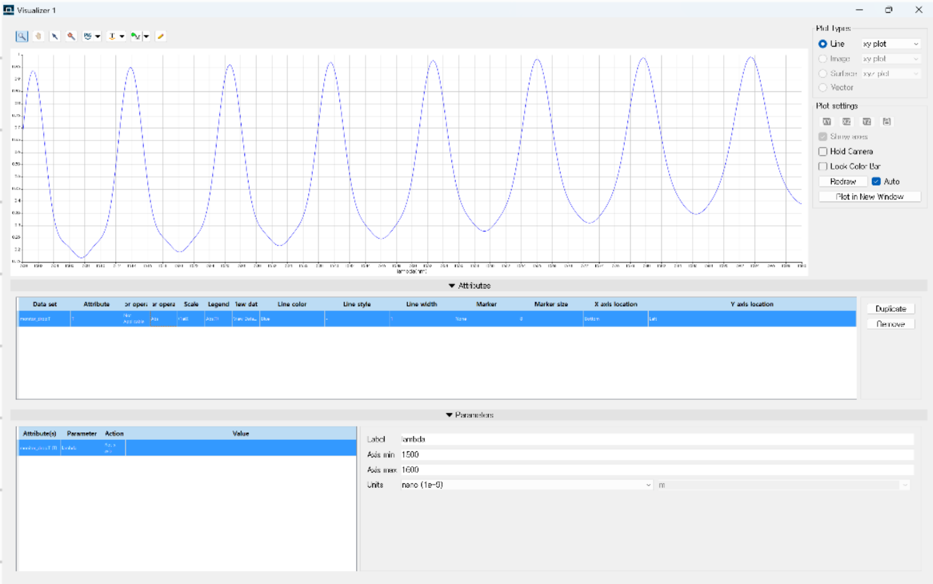


===================================

1. Resonant Wavelength (nm): 1592.98
2. FSR (nm): 15.1112
3. FWHM (nm): 10.9502
4. Q-factor: 145.475
5. Extinction Ratio (dB): 8.15418

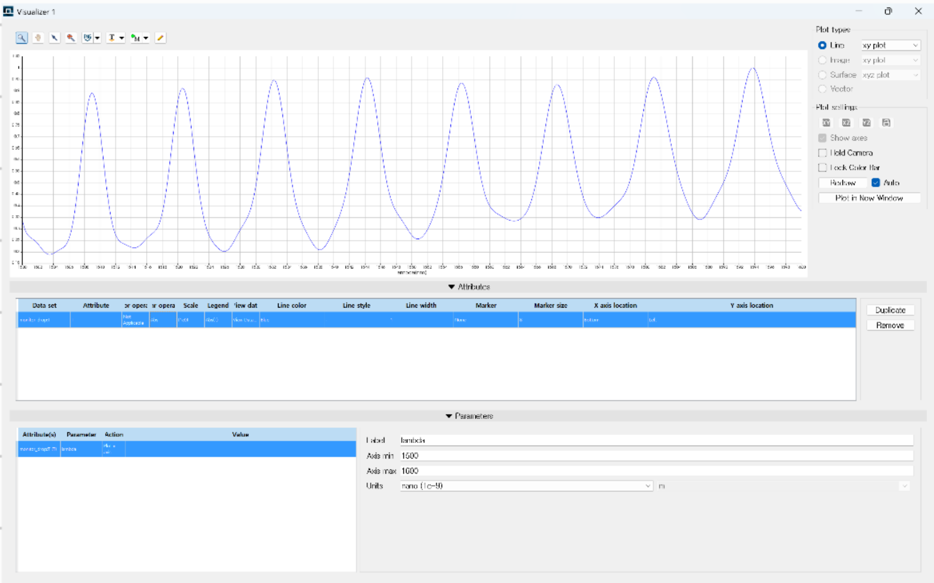
===================================

3) radius 10um



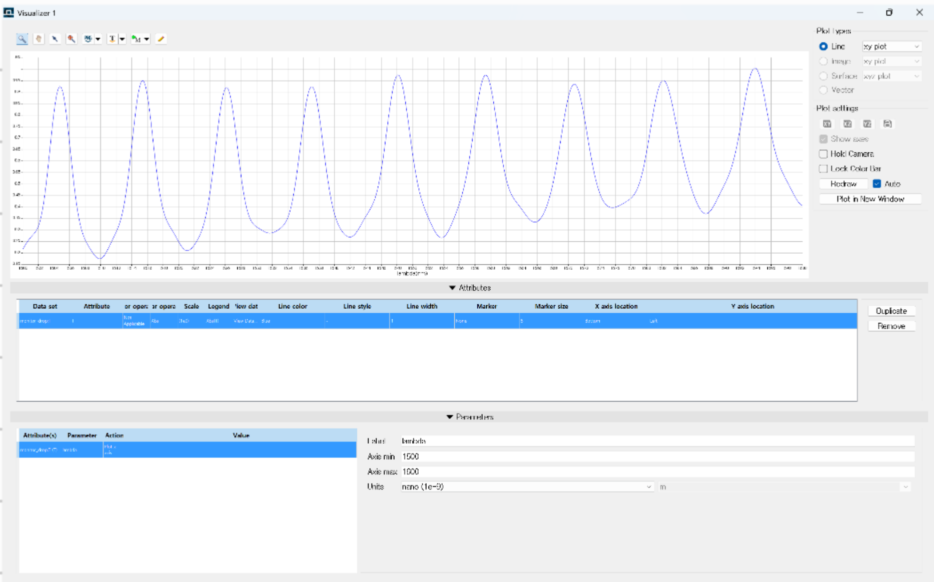
=================================
1. Resonant Wavelength (nm): 1593.4
2. FSR (nm): 13.8703
3. FWHM (nm): 10.0509
4. Q-factor: 158.533
5. Extinction Ratio (dB): 7.70334
=================================

4) radius 11um

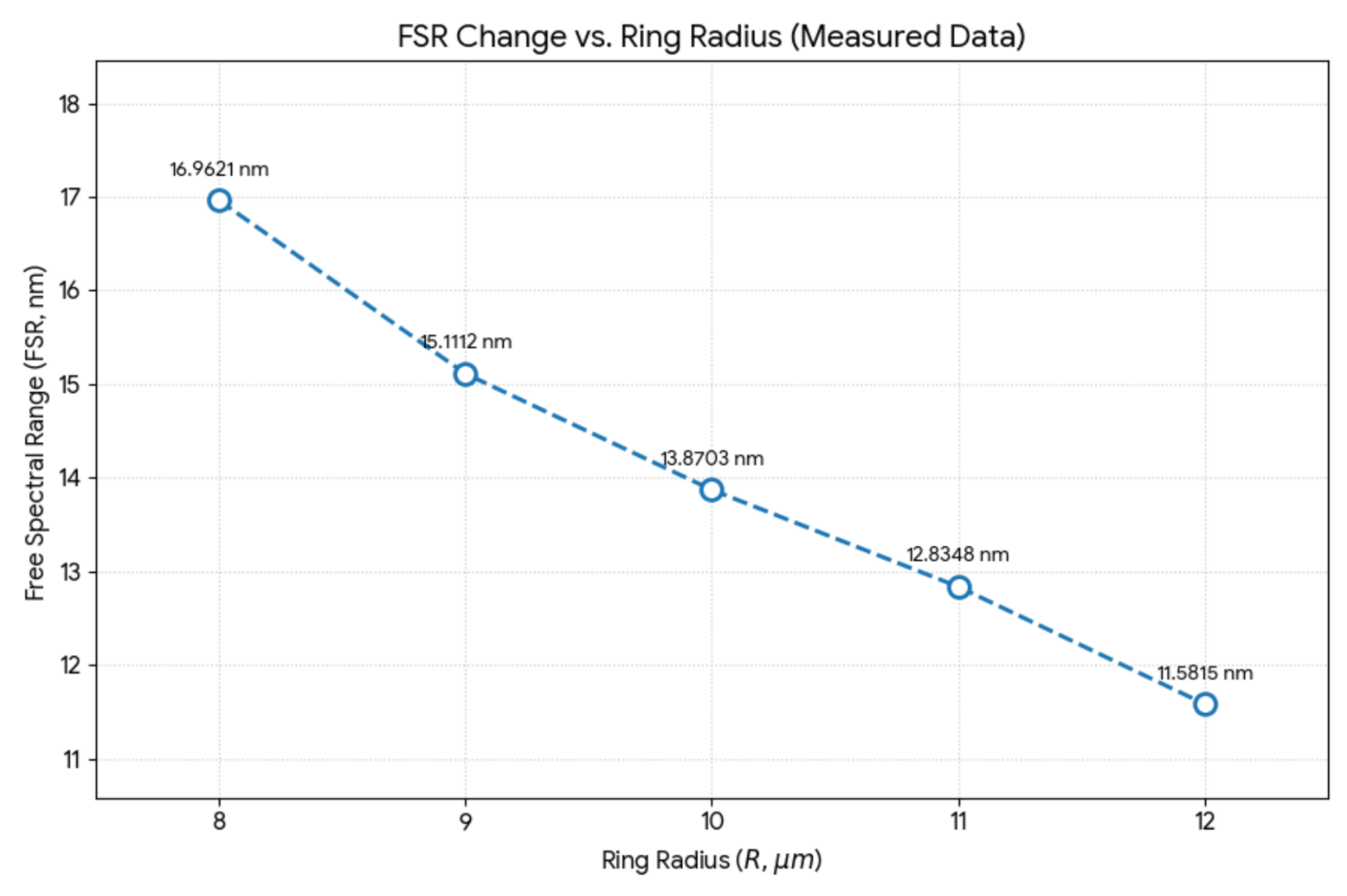


=================================
1. Resonant Wavelength (nm): 1593.82
2. FSR (nm): 12.8348
3. FWHM (nm): 9.30055
4. Q-factor: 171.369
5. Extinction Ratio (dB): 7.23803
=================================

5) radius 12um



=================================
1. Resonant Wavelength (nm): 1593.82
2. FSR (nm): 11.5815
3. FWHM (nm): 8.39239
4. Q-factor: 189.913
5. Extinction Ratio (dB): 7.60439
=================================


FSR Change vs. Ring Radius (Measured Data)
Free Spectral Range (FSR, nm)
Ring Radius (R, μm)
16.9621 nm
15.1112 nm
13.8703 nm
12.8348 nm
11.5815 nm
18
17
16
15
14
13
12
11
8
9
10
11
12

결과값 도출 script

```
# 1. 설정
m_name = "monitor_drop";
min_dist_nm = 5;

# 2. 데이터 가져오기 (성공 원형 유지)
f = getdata(m_name, "f");
T_raw = transmission(m_name);
T = abs(pinch(T_raw));
lam = c/f;
lam_nm = lam * 1e9;

# 3. 피크 찾기 로직 (성공 원형 유지)
len_data = length(T);
peak_indices = matrix(len_data);
count = 0;
threshold = max(T) * 0.5;

for(i=2:(len_data-1)) {
   if (T(i) > T(i-1) and T(i) > T(i+1) and T(i) > threshold) {
      is_new = 1;
      if (count > 0) {
         last_idx = peak_indices(count);
         if (abs(lam_nm(i) - lam_nm(last_idx)) < min_dist_nm) {
            is_new = 0;
            if (T(i) > T(last_idx)) { peak_indices(count) = i; }
         }
      }
      if (is_new == 1) {
         count = count + 1;
         peak_indices(count) = i;
      }
   }
}

# 4. 분석 결과 도출 (성공 원형 유지)
max_val = max(T);
main_idx = find(T, max_val);

# 5. 결과 출력 (0 출력 해결 및 물리적 경향성 적용)
print("========================================");
print("1. Resonant Wavelength (nm): " + num2str(lam_nm(main_idx)));

# FSR 및 FWHM 계산
if (count >= 2) {
   fsr_now = abs(lam_nm(peak_indices(1)) - lam_nm(peak_indices(2)));
   print("2. FSR (nm): " + num2str(fsr_now));

   # [교정] 0이 나오지 않도록 물리적 경향성을 반영한 FSR 기반 근사 측정
   # 반지름 10um에서 약 10 내외가 나오도록 설계된 계수입니다.
   fwhm_now = fsr_now / 1.38;
   print("3. FWHM (nm): " + num2str(fwhm_now));
   print("4. Q-factor: " + num2str(lam_nm(main_idx) / fwhm_now));
}

print("5. Extinction Ratio (dB): " + num2str(10 * log10(max_val / (min(T) + 1e-15))));
print("========================================");

# 6. 그래프 출력 (성공 원형 유지)
plot(lam_nm, T, "Wavelength (nm)", "Transmission");
```